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## chapter - 06

### 6.5 Exercises

(6.1) Here given,

Frequent closed itemsets = c

And support count of each of them = support(X)

In our algorithm we have to test a given itemset X is frequent or not.

So here X will be frequent, if x is a subset of a closed itemset, such as

$x \subseteq y$ (y is one of the c)

and, support(x) = support(y).

So the algorithm can be like

```
def check_itemset(x):
    for(each i in itemsets c):
        if{x ⊆ i and support(x) = support(i)}
            return support(x)
    return null
```

(6.2) (a) No, just using generators we can't determine support of all itemsets.

For example, if {A, B} generator has support count 5, we can't know the support count of {A}, as it can be greater

For this we need positive borders. Positive borders can be itemsets such that

1. It is frequent
2. It is not in generator
3. Its subset is " "

The algorithm can be defined as,

```
if (x in frequent or positive borders):
        return support(x)
else:
    for {x' ⊂ x and x' ∈ positive border}:
        if ∄ x'' = x' - {a} where x'' ∈ frequent
            and support(x'') = support(x')
                ∄ x = x - {a}
```

if (x is frequent or positive border):
    return support(x)
else
    return null

(b)

| Generator | Closed Itemset |
|---|---|
| 1. An itemset is generator if no proper subset has the same support. | 1. An itemset X is closed if no proper superset has the same support. |
| 2. It has smallest itemset that produces a particular support. | 2. It is the largest itemset that has a particular support. |

6.3 (a) Let's assume an itemset is frequent,

$$X = \{A, B, C\}$$

If $\{A, B, C\}$ appear in n transactions, then $\{A, B\}$ must appear in those as well

∴ So $\{A, B\}$ is also frequent.

(b) Let's assume,

$$s = \{A, B, C\}$$

and $$s' = \{A, B\}$$

So now every transaction containing $\{A, B, C\}$ must also has $\{A, B\}$.

But there might be some transactions where C is missing. So if we count then,

$$\text{sup\_count}(\{A, B\}) \geq \text{sup\_count}(\{A, B, C\})$$

or, $$\text{support}(\{A, B\}) \geq \text{support}(\{A, B, C\})$$

(c) Here, frequent itemset = l

and subset $s \subseteq l$

We have to proof,

$$c(s' \Rightarrow (l - s')) \leq c(s \Rightarrow (l - s))$$

Here, $s' \subseteq s$

Here, L.H.S. = $\text{confidence}(s' \Rightarrow (I - s'))$

$$= \frac{\text{support}(s' \cup (I - s'))}{\text{support}(s')}$$

$$= \frac{\text{support}(I)}{\text{support}(s')} \quad (\because \text{because } I \text{ contains everything})$$

R.H.S. = $\text{confidence}(s \Rightarrow (I - s))$

$$= \frac{\text{support}(s \cup (I - s))}{\text{support}(s)}$$

$$\geq \frac{\text{support}(I)}{\text{support}(s)}$$

We know, given is,

$$s' \subseteq s$$

So, $\text{support}(s') \geq \text{support}(s)$

or, $\frac{\text{support}(I)}{\text{support}(s')} \leq \frac{\text{support}(I)}{\text{support}(s)}$

or, L.H.S. $\leq$ R.H.S

[Proved]

(d) proof by contradiction:

Let's say, X is frequent in the D

$$\text{support}_D(x) \geq \text{minsup}$$

But X is not frequent in any partition.

$$\text{support}_{D_i}(x) < \text{minsup} \quad ; i \leq n$$

So, overall support will be,

$$\text{support}_D(x) = \sum_{i=1}^{n} \frac{N_i}{N} \text{support}_{D_i}(x)$$

Here,

$$\sum_{i=1}^{n} \frac{N_i}{N} = 1$$

As $\text{support}_{D_i}(x) < \text{minsup}$

there average will also be less,

$$\text{support}_D(x) < \text{minsup}$$

But this is false as we said in the beginning.

So, if x is frequent in D, then it's frequent in at least one partition.

(6.4) Here c is generated from $(k-1)$ frequent subsets. As those two were frequent, as per (our) rule, we do not need to check them again from consideration. So we only need to check $(k-2)$ items.

For example if

$l_1 = \{A, B, C\}$.

$l_2 = \{A, B, D\}$

and they combine into,

$c = \{A, B, C, D\}$

then we don't need to check $l_1$ and $l_2$

(6.5) From previous examples, we know that,

If $x \subseteq y$

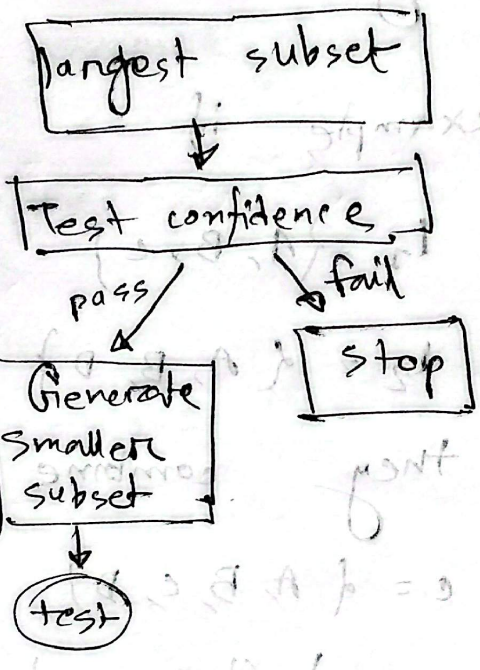
then support $(x) \geq$ support $(y)$

and confidence $(x \Rightarrow (l-x)) \leq$ confidence $(y \Rightarrow (1-y))$

So our idea is,

if $ABC \Rightarrow D$

and confidence $(ABC \Rightarrow D)$ fails then its subset AB, AC, BC will also fail




6.6 (a) Here given,

| TID | items brought |
|---|---|
| T100 | {M, O, N, K, E, Y} |
| T200 | {D, O, N, K, E, Y} |
| T300 | {M, A, K, E} |
| T400 | {M, U, C, K, Y} |
| T500 | {C, O, O, K, I, E} |

There are 5 transaction, so for being frequent appearance need $60\% \times 5 = 3$

For apriori,

$C_1$

| Itemset | Sup. Count |
|---|---|
| E | 4 |
| K | 5 |
| M | 3 |
| O | 3 |
| Y | 3 |
| N | 2 |
| D | 1 |
| A | 1 |
| U | 1 |
| C | 2 |
| I | 2 |

$L_1$

| Itemset | Sup |
|---|---|
| E | 4 |
| K | 5 |
| M | 3 |
| O | 3 |
| Y | 3 |

Now, generating $C_2$

| Itemset | sup. count |
|---|---|
| EK | 4 |
| EM | 2 |
| EO | 3 |
| EY | 2 |
| KM | 3 |
| KO | 3 |
| KY | 3 |
| MO | 1 |
| MY | 2 |
| OY | 2 |

$L_2$

| Itemset | Sup_count |
|---|---|
| EK | 4 |
| EO | 3 |
| KM | 3 |
| KO | 3 |
| KY | 3 |

Generating $C_3$

| Itemset | sup_count |
|---|---|
| EKO | 3 |

Generating $C_4$ isn't possible,

So the frequent itemsets are

{E}, {K}, {M}, {O}, {Y},
{E,K}, {E,O}, {K,M}, {K,O}, {K,Y},
{E,K,O}

For FP-Growth,

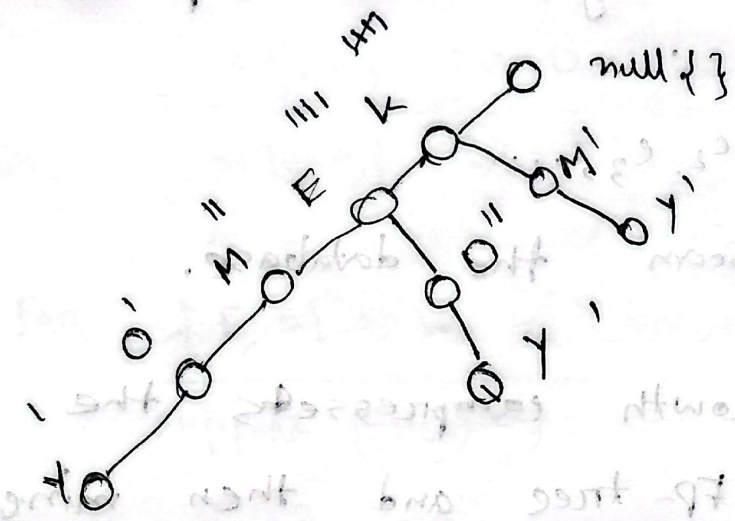
The $L_1$ will be,

| Itemset | sup_count |
|---|---|
| E | 4 |
| K | 5 |
| M | 3 |
| O | 3 |
| Y | 3 |

Arranging in descend suporder

K > E > M > O > Y

Now, by generating the tree we get,




| Item | Conditional Pattern Base | Conditional FP-tree | Pattern |
|---|---|---|---|
| Y | {kEMO:1, kEO:1, kM:1} | <k:3> <E:2> | {k,Y:3} |
| | | ~~<M:2> <O:2>~~ | {k,O:3} |
| O | {kEM:1, kE:2} | <k:3><E:3> | {E,O:3}, {Ek,O:3} |
| M | {kE:2, k:1} | ~~<k:3><E:2>~~ | {k,M:3} |
| E | {k:4} | <k:4> | {E,k:4} |

FP Growth result

{E}, {k}, {M}, {O}, {Y}

{k,Y}, {E,k,O}, {k,O},

{E,O}, {k,M}, {E,k}

(6.8) (a) the updated list of transactions,

| cust_id | TID | items_bought |
|---|---|---|
| 1 | T100 | {Crab, Milk, Cheese, Bread} |
| 2 | T200 | {Cheese, Milk, Apple, Pie, Bread} |
| 1 | T300 | {Apple, Milk, Bread, Pie} |
| 3 | T400 | {Bread, Milk, Cheese} |

Using apriory,

$C_1$

| TID | items |
|---|---|
| Bread | 4 |
| Milk | 4 |
| Cheese | 3 |
| Apple | 2 |
| Pie | 2 |
| Crab | 1 |

$T_1$

| TID | Items |
|---|---|
| Bread | 4 |
| Milk | 4 |
| Cheese | 3 |

there minimum support count $4 \times 60\% \approx 3$

Then, $C_2$

| TID | items |
|---|---|
| Bread, Milk | 4 |
| Bread, cheese | 3 |
| Milk, cheese | 3 |

And for frequent 3 itemsets,

| TID | items | |
|---|---|---|
| | Bread, Milk, Cheese | 3 |

As per the question,

$item_1 \wedge item_2 \Rightarrow item_3$

we need 3 itemsets, Here,

Bread ∧ cheese ⇒ Milk

support, $s = \frac{3}{4} = 75\%$

confidence, $c = \frac{3}{3} = 100\%$

cheese ∧ Milk ⇒ Bread

confidence, $c = \frac{3}{3} = 100\%$

Bread ∧ Milk ⇒ Cheese

confidence, $c = \frac{3}{4} = 75\%$

This is not strong $75\% < 80\%$

So, we can take cheese ⇒ Bread ∧ Milk

(b) Form brand-item category.

| cust_id | items_bought |
|---|---|
| 1 | {king's_crab, sunset-milk, Dairyland-cheese, Best-bread, Westcoast-Apple, Dairyland-Milk, Wonder-Bread, Tasty-Pie} |
| 2 | {Best-cheese, Dairyland-Milk, Goldenfarm-Apple, Tasty-Pie, Wonder-Bread} |
| 3 | {Wonder-Bread, Sunset-Milk, Dairyland-Cheese} |

So the minimum support count = $3 \times 60\% \approx 2$

Here, $C_1$

| cust_id | freq_count |
|---|---|
| Wonder-Bread | 3 |
| Dairyland-Milk | 2 |
| Tasty-Pie | 2 |
| Sunset-Milk | 2 |
| Dairyland-cheese | 2 |
| King's Crab | 1 |
| BestBread | 1 |
| WestCoastApple | 1 |
| BestCheese | 1 |
| GoldenFarmApple | 1 |

$L_1$

| cust_id | sup_count |
|---|---|
| WonderBread | 3 |
| DairylandMilk | 2 |
| TastyPie | 2 |
| SunsetMilk | 2 |
| DairylandCheese | 2 |

Generating $C_2$,

| Cust_id | sup_count |
|---|---|
| WonderBread, DairylandMilk | 2 |
| WonderBread, TastyPie | 2 |
| WonderBread, SunsetMilk | 2 |
| WonderBread, DairylandCheese | 2 |
| DairylandMilk, TastyPie | 2 |
| DairylandMilk, SunsetMilk | 1 |
| DairylandMilk, DairylandCheese | 1 |
| TastyPie, SunsetMilk | 1 |
| TastyPie, DairylandCheese | 1 |
| SunsetMilk, DairylandCheese | 2 |

And after pruning $L_3$

| Cust_id | sup_count |
|---|---|
| WonderBread, DairylandMilk, TastyPie | 2 |
| WonderBread, SunsetMilk, DairylandCheese | 2 |

(6.9) Here, if we consider 4 shops, then,

global support count

$$Count_D(X) = count_1(X) + \cdots + count_4(X)$$

So we can mine locally, and then each site will send locally frequent itemsets and their support counts over the network.

And afterwards we can derive strong association rules from the global frequent itemsets.

(6.10) Let DB be the original transactional database. Let $\Delta DB$ the newly added transactions. So the new database

$$DB' = DB + \Delta DB$$

P.T.O.

Here instead of mining from scratch, we can do the following:

1. Store the frequent itemsets and their support counts from the original dataset. It is also useful to set negative border.
2. Construct a candidate set from the old frequent itemsets and the relevant negative border itemsets.
3. Scan only the newly added transactions and count their occurances.
4. Update the support count, using,

$$count_{DB'}(X) = count_{DB}(X) + count_{ADB}(X)$$

5. Compute the new support using,

$$support_{DB'}(X) = \frac{count_{DB'}(X)}{|DB| + |\Delta DB|}$$

6. If any candidate satisfies the minimum-support threshold, then that becomes frequent.
7. Finally generate association rules.

6.11 Let's say for example,

a transaction,

$T_1 = \{cake, cake, Milk\}$

which is treated as $\{cake, Milk\}$

But we want to preserve $\{cake:2, Milk:1\}$

So we can use quantity-aware mining. The idea is to treat quantity as a part of the item.

So above data will look like,

$\{cake_1, cake_2, Milk\}$

Here, support $(cake_1)$, support $(cake_2)$ will be used instead of single cake.

Likewise, for FP-Growth we can store quantity alongside, instead of creating duplicate nodes for efficiency.

6.13 Let's consider,

1000 total transition

Sunglass count = 900

Umbrella count = 100

Both sunglass & umbrella = 80

So if we use the rule

Umbrella → Sunglass

confidence $\frac{80}{100} = 80\%$

If the minimum confidence is 70%, it is strong

But, if we look at probability,

$P(\text{sunglass}) = \frac{960}{1000} = 90\%$

But $P(\text{Sunglass} \mid \text{Umbrella}) = \frac{80}{100} = 80\%$

Here, $P(\text{sunglass} \mid \text{umbrella}) < P(\text{sunglasses})$

So they are negatively correlated

6.14 (a) Here the given table

| | Hot dog | not dog | |
|---|---|---|---|
| hamburgers | 2000 | 500 | 2500 |
| not hamburgers | 1000 | 1500 | 2500 |
| | 3000 | 2000 | 5000 |

Given rule,

hot dogs $\Rightarrow$ hamburger

Here, support $= \frac{2000}{5000} = 40\%$

confidence $\frac{2000}{3000} = 66.7\%$

They meet the requirements,

so the rule is strong.

(b) To learn the correlation between hotdog and hamburger

hotdog ⇒ hamburger

$$\text{Corr}_{\{\text{hotdog, hamburger}\}} = \frac{P\{\text{hotdog, hamburger}\}}{P\{\text{hotdog}\}\, P\{\text{hamburger}\}}$$

$$= \frac{0.4}{0.5 \times 0.6}$$

$$= 1.33 > 1$$

So they are not independent, a positive correlation exists.

(c) Let's first calculate,

all-confidence

max-confidence

kulczynski

cosine measures